A $10 Chip That Could Put a Lab on Your Wrist

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A **spectrometer** is a tool that studies light to learn what things are made of. Labs use big, costly ones to test medicines, foods and chemicals. Now, researchers at the University of Cambridge have built one for about $10. It is only a few centimeters wide small enough, one day, to fit in a health gadget worn on the wrist.

The team calls it a **convolutional** spectrometer**. It works on the light directly, then uses quick math to work out the** spectrum**. In tests, it correctly named plastics, medicines, coffee, flour and tea every single time. It also measured** health markers **such as** blood sugar **without needles, and it kept working from very cold to very hot.**

The researchers warn that much more testing is needed before it can become a real medical product. Still, it could lead to cheap, exact **sensors** for health monitors and food testing.

Word Treasure

spectrometer -- A tool that studies light to learn what things are made of.

convolutional spectrometer -- The team's new, tiny kind of spectrometer that uses quick math to study light.

spectrum -- The pattern of light that helps scientists tell what something is made of.

health markers -- Signs in your body, like blood sugar, that show how healthy you are.

blood sugar -- The amount of sugar in your blood.

sensors -- Devices that notice and measure things like light, heat or movement.

Background you need

A spectrometer is a lab tool that studies light to figure out what things are made of.

Until now, these tools were big and costly, so they mostly stayed inside labs, where scientists use them to test medicines, foods and chemicals.

If the same tool could be made tiny and cheap, it could fit inside everyday gadgets, like a health monitor worn on the wrist.

Break it down

WHO: Researchers at the University of Cambridge

WHAT: They built a spectrometer for about $10 that is only a few centimeters wide and uses quick math to work out the spectrum

WHERE: At the University of Cambridge, tested on plastics, medicines, coffee, flour and tea

WHY: So that testing medicines, foods and health markers could one day be done by small, cheap sensors instead of big, costly lab machines

Why it matters

One day, a gadget on your wrist might check things like your blood sugar without needles and help make sure the food you eat is safe.

Test what you remember

1. What does a spectrometer do?

- (A) It cooks food with light
- (B) It takes photos of stars
- (C) It measures how loud sounds are
- (D) It studies light to learn what things are made of

2. Who built the new tiny spectrometer?

- (A) Astronauts in space
- (B) A toy company
- (C) Researchers at the University of Cambridge
- (D) Doctors at a hospital

3. About how much does the new spectrometer cost to build?

- (A) About $500
- (B) About $10,000
- (C) It is free
- (D) About $10

4. In tests, what did the chip correctly name every single time?

- (A) Plastics, medicines, coffee, flour and tea
- (B) Types of clouds
- (C) Songs played on the radio
- (D) Colors of paint

5. How did the chip measure blood sugar?

- (A) Without needles
- (B) By asking questions
- (C) With a tiny camera
- (D) With a thermometer

6. What do the researchers warn about their invention?

- (A) It only works in the cold
- (B) It costs too much to make
- (C) Much more testing is needed before it can become a real medical product
- (D) It is too big to wear

Answer key (try the quiz first!): 1: D 2: C 3: D 4: A 5: A 6: C

Questions to think about

1. The researchers: They warn that much more testing is needed before the chip can become a real medical product.

MY THOUGHTS (at least 20 words):